

A Counterexample to Conjecture 00000008547

gaochengzhecpu (AI-assisted submission)

The assertion that every geometric Peck lattice has a symmetric characteristic polynomial is false. The five-element diamond lattice M_3 is a counterexample: it is geometric, Peck, and strictly Sperner, but its characteristic polynomial is

$$\chi_{M_3}(t) = t^2 - 3t + 2.$$

Its constant and leading coefficients differ. Thus its coefficients are not symmetric about half its rank, which is 1.

Conventions and scope. A finite lattice is geometric if it is atomistic and upper semimodular. We use the standard Peck conditions: rank symmetry, rank unimodality, and the strong Sperner property [1]. The source instead says “rank-symmetric strictly Sperner.” Our example satisfies this condition too: its unique maximum antichain is its middle rank. We disprove the universal characteristic-polynomial assertion, which is one conjunct of the conjecture; no claim about the infinite Young lattice is needed.

The lattice and its geometric property. Let

$$M_3 = \{0, a, b, c, 1\}, \quad 0 < a < 1, \quad 0 < b < 1, \quad 0 < c < 1,$$

with a, b, c pairwise incomparable. Two distinct atoms have meet 0 and join 1; all other meets and joins are forced by comparability. Every element is the join of the atoms below it, using the empty join for 0, so the lattice is atomistic. Its rank function has values 0, 1, 1, 1, 2.

To verify upper semimodularity, suppose x covers $x \wedge y$. If $x \leq y$, the premise is impossible. If $y < x$, then $x \wedge y = y$ and $x \vee y = x$, so the conclusion repeats the premise. The only remaining case consists of two distinct atoms. Their meet is 0, their join is 1, and 1 covers y . Hence M_3 is geometric.

Both Sperner requirements. The rank sizes are $(1, 3, 1)$, so they are symmetric and unimodal. Any antichain containing 0 or 1 is a singleton; every other antichain is a subset of $\{a, b, c\}$. Consequently the largest antichain has size 3, and it is uniquely the middle rank. In particular the lattice is strictly Sperner.

A union of two antichains cannot contain all five elements: the chain $0 < a < 1$ would then contribute three comparable elements to two antichains. The upper bound 4 is attained by $\{0, a, b, c\}$, the union of the bottom and middle ranks. A union of three antichains has size at most 5, attained by the whole lattice. For more antichains the same bound is attained by allowing empty antichains. These bounds are exactly the sums of the largest k rank sizes. Thus M_3 is strongly Sperner, and hence Peck also in the standard sense.

The actual characteristic polynomial. Write $\mu(x) = \mu(0, x)$. The defining Möbius recurrence is

$$\sum_{y \leq x} \mu(y) = \begin{cases} 1 & x = 0, \\ 0 & x \neq 0. \end{cases}$$

It gives $\mu(0) = 1$, $\mu(a) = \mu(b) = \mu(c) = -1$, and $\mu(1) = -(1 - 1 - 1 - 1) = 2$. Therefore the defining sum gives

$$\chi_{M_3}(t) = \sum_{x \in M_3} \mu(x) t^{2-\text{rk}(x)} = t^2 - 3t + 2.$$

The coefficient sequence in ascending degree is $(2, -3, 1)$, so it is not palindromic. If “symmetric” were instead intended as a reflection of values or roots about 1, that also fails: $\chi(2) = 0$ whereas $\chi(0) = 2$.

Formal verification and exhaustive scope. The accompanying Lean 4.19.0 file uses only `Std`. Its type `LatticeData` includes the order, bottom, top, meet, join, and rank on five elements. It checks the partial-order and lattice laws, the cover-based rank rule, atomisticity, and upper semimodularity.

Every subset is encoded by its five membership bits, as an element of `Fin 32`. The theorem `every_prop_subset_encoded` covers arbitrary proposition-valued subsets as well. Chains and antichains use the actual order, and a k -family is defined by the absence of a chain of more than k elements. Such families include all unions of k antichains: each chain meets any one antichain in at most one element. Lean proves the exact extremal bound, with an attaining family, for every natural number k ; no list of specially chosen families replaces the universal quantifier. The unique maximum antichain is also checked.

The Möbius values satisfy the defining recurrence, and `diamond_mobius_unique` proves that every solution of that recurrence is this one. The coefficients are sums of these values over the appropriate ranks, rather than an unrelated polynomial declared by hand. Finally, `conjecture_8547_false` negates the universal symmetry assertion specialized to five-element lattices, even with both Peck and strict Sperner imposed as hypotheses. This specialization is necessary for the conjecture as stated, so its failure disproves the conjecture. There are no admitted proofs or additional axioms; the final theorem uses only `propext` and `Quot.sound`.

References

- [1] R. P. Stanley, *Quotients of Peck posets*, Order 1 (1984), 29–34, p. 29. <https://math.mit.edu/~rstan/pubs/pubfiles/60.pdf>